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Business Requirements Document: Analytical Data Platform Initiative

Cover Page

ANALYTICAL DATA PLATFORM INITIATIVE

Separating Transactional and Analytical Workloads to Improve Decision Velocity and Operational Efficiency

Prepared for: Executive Steering Committee

Prepared by: Chief Technology Officer, Chief Financial Officer, VP of Data & Analytics

Date: June 18, 2026

Status: For Review and Approval

Recommendation: Proceed with implementation

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1. Executive Summary

1.1 Situation

Our company operates a high-volume e-commerce platform processing approximately 50,000 orders per day. The transactional database (PostgreSQL) is currently serving dual purposes: operational transaction processing and analytical query workloads. This architectural mismatch is creating measurable business friction.

1.2 Problem

Analytical queries against the transactional database are causing:

- **Slow insights:** Average query latency of 5-30 minutes, delaying operational decisions
- **Peak-load fragility:** Analytics queries lock resources during peak traffic, degrading customer-facing performance
- **High operational toil:** Data engineering team spends 60-70% of capacity on maintenance and incident response rather than building new capabilities
- **Limited scalability:** The system cannot support real-time analytical use cases

1.3 Proposed Solution

Implement a purpose-built analytical platform that separates transactional and analytical workloads. This platform will:

- Extract data from the transactional system incrementally
- Transform and materialize data in a high-performance analytical engine
- Expose analytical capabilities via natural language interfaces
- Monitor system health autonomously

1.4 Business Case Summary

Metric	Year 1	Year 2	Year 3
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Incremental Revenue	\$2.1M	\$3.8M	\$5.2M
Cost Reduction	\$385K	\$425K	\$450K
Total Benefit	\$2.485M	\$4.225M	\$5.65M
Investment	\$140K (dev) + \$185K (ops)	\$210K (ops)	\$230K (ops)
Net Benefit	\$2.16M	\$4.015M	\$5.42M
Cumulative ROI	78%	195%	352%
Payback Period	10.2 months	—	—

1.5 Recommendation

Proceed with implementation. The initiative delivers positive ROI within 12 months, addresses a material operational constraint, and enables future analytical capabilities. The cost of inaction exceeds the cost of implementation by a factor of 3:1 over three years.

2. Current State Assessment

2.1 Operational Context

Company Profile

- Mid-size e-commerce operator
- ~\$500M annual GMV (Gross Merchandise Value)
- 50,000 orders/day average; 75,000 orders/day peak
- Average order value: \$85
- Gross margin: 28%
- Data team: 8 full-time engineers

Current Technology Stack

- PostgreSQL 14 (transactional database, 2TB, 3-node cluster)
- Custom Python ETL scripts (cron-based, daily runs)
- Tableau (BI layer, ~30 dashboards, 50 active users)
- Manual incident response (on-call rotation, 2-3 incidents/month)

2.2 Current Analytics Performance

Metric	Current	Target	Gap
Query latency (p95)	12 minutes	<5 seconds	144x slower
Query latency (p99)	28 minutes	<15 seconds	112x slower
Data freshness	24 hours	Real-time	24-hour lag
System availability	97.8%	99.5%	1.7% gap
Incident MTTR	5.2 hours	30 minutes	10x slower
Incidents/month	2.8	<0.5	5.6x higher

Data Source: Operational logs (Jan-May 2026); query performance monitoring; incident tracking system

2.3 Current Cost Structure

Annual Spend (Status Quo)

Category	Amount	Notes
Infrastructure		
PostgreSQL cluster (3-node, 2TB)	\$145K	On-premises + cloud hybrid
Storage & backup	\$28K	Redundant storage, daily backups
Personnel		
Data engineers (8 FTE @ \$145K loaded)	\$1,160K	Includes benefits, overhead
Maintenance & incident response (est. 60% of capacity)	\$696K	Derived from time tracking
Operational Costs		
On-call rotation stipends	\$32K	2-3 incidents/month, \$400/incident
Emergency vendor support	\$18K	Ad-hoc performance tuning
Total Annual Cost	\$2,079K	

3. Problem Statement & Business Impact

3.1 Core Problem

The transactional database is being used for analytical workloads it was not designed to support. This creates a cascading set of business problems.

3.2 Impact 1: Slow Decision-Making

Symptom: Analytical queries take 5-30 minutes to execute.

Business Impact: Operational decisions that could be made in real-time are delayed by hours or days.

Quantified Example:

- Pricing optimization: Currently runs once per day (overnight batch). Competitors with real-time analytics adjust prices every 15-30 minutes based on demand signals.
- During a 10-hour peak sales window, a competitor can make 20-40 pricing adjustments; we make 0-1.
- Estimated revenue loss per peak event: 3-5% of peak-window GMV
- Peak-window GMV (Black Friday, holiday season): ~\$2M per event
- Estimated loss per event: \$60-100K
- Frequency: ~8 major peak events per year
- **Annual revenue impact: \$480K-800K** (Confidence: Medium)

Requirement ID: BR-001 - Reduce analytical query latency to <5 seconds (p95)

3.3 Impact 2: Peak-Load Fragility

Symptom: During peak traffic, analytics queries lock database resources, degrading customer-facing performance.

Business Impact: Reduced conversion rates and customer experience during high-value periods.

Quantified Example:

- During peak traffic (Black Friday, Cyber Monday, holiday season), heavy analytics queries occasionally lock tables, causing transaction processing to slow.
- Estimated impact: 0.5-1% conversion rate degradation during 2-3 peak events per year

- Annual GMV: \$500M; peak-window GMV: \$2M per event
- Conversion rate impact: $0.5-1\% \times \$2M \times 8 \text{ events} = \$80-160K$ revenue loss
- **Annual revenue impact: \$80-160K** (Confidence: Medium)

Requirement ID: BR-002 - Isolate analytical workloads from transactional workloads

3.4 Impact 3: High Operational Toil

Symptom: Data engineers spend 60-70% of time maintaining the system rather than building new capabilities.

Business Impact: Slower time-to-market for new analytical features; reduced engineering velocity.

Quantified Example:

- Data team: 8 FTE @ \$145K fully-loaded cost = \$1.16M annual personnel cost
- Estimated maintenance burden: 60-70% of capacity = \$696-812K annually
- This represents 4.8-5.6 FTE spent on maintenance
- Opportunity cost: 2.4-3.2 FTE could be building new capabilities instead

New Capabilities Foregone:

- Real-time personalization engine: Could generate +\$300-500K annual revenue
- Predictive inventory optimization: Could reduce inventory carrying costs by \$150-250K annually
- Churn prediction model: Could improve retention by 1-2%, worth \$200-400K annually
- **Total opportunity cost: \$650K-1.15M annually** (Confidence: Medium)

Requirement ID: BR-003 - Reduce data engineering maintenance burden to <30% of capacity

3.5 Impact 4: Limited Scalability

Symptom: The system cannot support real-time analytical use cases.

Business Impact: Cannot compete on data-driven features; competitive disadvantage.

Quantified Example:

- Real-time personalization (showing personalized recommendations on every page view) is not feasible with 24-hour data freshness
- Competitors with real-time analytics report 15-20% higher average order value through personalization

- Our current AOV: \$85
- Potential uplift: $\$85 \times 15\text{-}20\% = \$12.75\text{-}17$ per order
- Daily orders: 50,000
- **Potential annual revenue uplift: \$233M-310M** (Confidence: Low - depends on implementation quality and market conditions)
- **Conservative estimate (5% of potential): \$11.6-15.5M annually** (Confidence: Medium)

Requirement ID: BR-004 - Enable real-time analytical capabilities

3.6 Summary: Cost of Inaction

If no action is taken, the company will incur the following costs over 3 years:

Cost Category	Year 1	Year 2	Year 3	3-Year Total
Pricing optimization delay	\$640K	\$700K	\$760K	\$2.1M
Peak-load fragility	\$120K	\$140K	\$160K	\$420K
Opportunity cost (foregone capabilities)	\$850K	\$950K	\$1.05M	\$2.85M
Continued maintenance burden	\$696K	\$750K	\$810K	\$2.256M
Total Cost of Inaction	\$2.306M	\$2.54M	\$2.78M	\$7.626M

Baseline: Doing nothing costs \$7.6M over three years. The proposed investment must be judged against this baseline.

4. Proposed Solution & Capabilities

4.1 Solution Overview

Implement a purpose-built analytical platform that separates transactional and analytical workloads. The platform will extract data from the transactional system, transform it for analytical use, and expose it via natural language interfaces.

4.2 Functional Capabilities

Capability 1: Incremental Data Extraction

- Extract new and changed records from the transactional database incrementally (not full daily dumps)
- Reduce data movement by 80-90% compared to current approach
- Enable near-real-time data freshness (minutes instead of hours)
- **Requirement ID:** BR-005

Capability 2: Data Transformation & Materialization

- Apply business logic transformations (cleansing, deduplication, enrichment)
- Materialize data in a high-performance analytical engine
- Support complex analytical queries without impacting transactional performance
- **Requirement ID:** BR-006

Capability 3: Natural Language Query Interface

- Enable business users to query analytical data using natural language
- Reduce dependency on data engineers for ad-hoc reporting
- Support both structured queries and exploratory analysis
- **Requirement ID:** BR-007

Capability 4: Autonomous System Monitoring

- Detect anomalies and failures in the analytical pipeline
- Diagnose root causes without human intervention
- Propose and execute remediation steps
- Generate post-incident reports
- **Requirement ID:** BR-008

4.3 Non-Functional Requirements

Requirement	Target	Rationale
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Query latency (p95)	<5 seconds	Enable real-time decision-making
Query latency (p99)	<15 seconds	Acceptable for exploratory analysis
Data freshness	<5 minutes	Support near-real-time use cases
System availability	99.5%	Acceptable for analytical workloads
Incident MTTR	<30 minutes	Minimize business impact
Cost per query	<\$0.10	Economically viable at scale

Requirement ID: BR-009

4.4 Out of Scope

The following are explicitly out of scope for this initiative:

- Real-time operational dashboards (separate initiative)
- Advanced machine learning pipelines (Phase 2)
- Data governance and compliance enhancements (separate initiative)
- Migration of historical data beyond 12 months (Phase 2)

5. Financial Analysis

5.1 Revenue Impact Analysis

5.1.1 Pricing Optimization (BR-001)

Current State:

- Pricing optimization runs once per day (overnight batch)
- Average pricing adjustment: 2-3% margin improvement
- Current annual margin: \$500M GMV × 28% = \$140M

With Analytical Platform:

- Pricing optimization can run every 15-30 minutes during peak periods
- Estimated additional margin improvement: 1-2% (incremental to current 2-3%)

- Applies to: Peak-window GMV only (~\$50M annually, 10% of total GMV)

Calculation:

- Incremental margin: $\$50M \times 1-2\% = \$500K-1M$
- Confidence: Medium (depends on pricing algorithm effectiveness)
- Conservative estimate: \$500K annually
- **Year 1 Revenue Impact: \$500K** (Confidence: Medium)

5.1.2 Peak-Load Isolation (BR-002)

Current State:

- Peak-window conversion rate: 2.8%
- Peak-window GMV: ~\$2M per event, ~8 events per year
- Occasional analytics-induced slowdowns reduce conversion by 0.5-1%

With Analytical Platform:

- Analytical workloads isolated from transactional workloads
- Conversion rate improvement: 0.3-0.5% (conservative estimate of recoverable loss)

Calculation:

- Peak-window GMV: $\$2M \times 8 \text{ events} = \$16M$ annually
- Conversion rate improvement: $0.3-0.5\% \times \$16M = \$48-80K$
- Confidence: Medium
- **Year 1 Revenue Impact: \$50K** (Confidence: Medium)

5.1.3 Real-Time Personalization (BR-004)

Current State:

- Personalization based on 24-hour-old data
- Current AOV: \$85
- Daily orders: 50,000

With Analytical Platform:

- Real-time personalization becomes feasible
- Estimated AOV uplift: 2-4% (conservative; competitors report 5-8%)
- Applies to: 30% of orders (remaining 70% are repeat customers with existing preferences)

Calculation:

- Addressable orders: $50,000 \times 30\% = 15,000$ per day
- AOV uplift: $\$85 \times 2\text{-}4\% = \$1.70\text{-}3.40$ per order
- Daily incremental revenue: $15,000 \times \$1.70\text{-}3.40 = \$25.5\text{-}51\text{K}$
- Annual incremental revenue: $\$25.5\text{-}51\text{K} \times 365 = \$9.3\text{-}18.6\text{M}$
- Confidence: Low (depends on implementation quality, market conditions)
- Conservative estimate (50% of low-end): $\$4.65\text{M}$ annually
- **Year 1 Revenue Impact: \$1.05M** (Confidence: Low)

Note: Real-time personalization requires additional development beyond the core analytical platform. This estimate assumes Phase 2 implementation.

5.1.4 Faster Time-to-Market for New Capabilities (BR-003)

Current State:

- New analytical capability: 3-4 weeks to implement
- Data team capacity: 8 FTE, 60-70% on maintenance = 2.4-3.2 FTE available for new work

With Analytical Platform:

- New analytical capability: 1-2 weeks to implement (50% faster)
- Data team capacity: 8 FTE, 20-30% on maintenance = 5.6-6.4 FTE available for new work
- Additional capacity: 3.2-4 FTE

Capabilities Enabled:

- Inventory optimization: $\$150\text{-}250\text{K}$ annual value
- Churn prediction: $\$100\text{-}200\text{K}$ annual value
- Demand forecasting: $\$50\text{-}100\text{K}$ annual value
- **Total: \$300-550K annually**
- Conservative estimate (50% of low-end): $\$150\text{K}$ annually
- **Year 1 Revenue Impact: \$550K** (Confidence: Medium)

5.1.5 Total Year 1 Revenue Impact

Driver	Amount	Confidence
Pricing optimization	\$500K	Medium

Peak-load isolation	\$50K	Medium
Real-time personalization (Phase 2)	\$1.05M	Low
Faster time-to-market	\$550K	Medium
Total	\$2.15M	Medium

Conservative Estimate (50% of low-confidence items): \$2.1M

Confidence Level: Medium

5.2 Cost Reduction Analysis

5.2.1 Reduced Maintenance Burden (BR-003)

Current State:

- Data team: 8 FTE @ \$145K fully-loaded = \$1.16M annually
- Maintenance burden: 60-70% of capacity
- Maintenance cost: \$696-812K annually

With Analytical Platform:

- Maintenance burden: 20-30% of capacity
- Maintenance cost: \$232-348K annually
- Reduction: \$348-580K annually

Calculation:

- Current maintenance: \$750K (midpoint)
- Future maintenance: \$290K (midpoint)
- Savings: \$460K annually
- Confidence: Medium (depends on platform reliability)
- Conservative estimate (80% of savings): \$368K annually
- **Year 1 Cost Reduction: \$368K** (Confidence: Medium)

5.2.2 Reduced Incident Response Costs

Current State:

- Incidents per month: 2.8

- Average incident duration: 5.2 hours
- Cost per incident: \$400 (on-call stipend + emergency support)
- Annual incident cost: $2.8 \times 12 \times \$400 = \13.4K

With Analytical Platform:

- Autonomous monitoring reduces incidents by 70-80%
- Incidents per month: 0.6-0.8
- Cost per incident: \$200 (reduced severity, faster resolution)
- Annual incident cost: $0.7 \times 12 \times \$200 = \1.7K
- Savings: \$11.7K annually

Calculation:

- Also includes reduced on-call burden: $2 \text{ FTE} \times \$32\text{K} = \64K annually
- Total on-call savings: $\$11.7\text{K} + \$64\text{K} = \$75.7\text{K}$
- Conservative estimate: \$17K annually
- **Year 1 Cost Reduction: \$17K** (Confidence: Medium)

5.2.3 Total Year 1 Cost Reduction

Driver	Amount	Confidence
Reduced maintenance burden	\$368K	Medium
Reduced incident response	\$17K	Medium
Total	\$385K	Medium

5.3 Investment Required

5.3.1 Development Investment (One-Time)

Category	Cost	Notes
Lead Data Engineer (4 months)	\$48K	$\$145\text{K annual} \times 4/12$
Agentic Engineer (4 months)	\$48K	$\$145\text{K annual} \times 4/12$
Backend Engineer (2 months)	\$24K	$\$145\text{K annual} \times 2/12$

Total Personnel	\$120K	
External services & tools	\$20K	Licenses, vendor setup, consulting
Total Development	\$140K	

Confidence: High (based on comparable projects)

5.3.2 Operational Investment (Annual)

Category	Year 1	Year 2	Year 3	Notes
Infrastructure				
Analytical database service	\$45K	\$55K	\$65K	Compute-based pricing, scales with usage
Orchestration platform	\$25K	\$30K	\$35K	Serverless tier
API hosting	\$8K	\$10K	\$12K	Cloud Run/Lambda
Personnel				
Platform maintenance (0.5 FTE)	\$73K	\$75K	\$77K	\$145K annual × 0.5
LLM API Costs				
Autonomous monitoring	\$34K	\$42K	\$50K	GPT-4o for manager, GPT-4o-mini for agents
Total Annual	\$185K	\$212K	\$239K	

Confidence: Medium (LLM costs depend on usage; infrastructure costs scale with data volume)

5.4 Financial Summary: Base Case

Metric	Year 1	Year 2	Year 3	3-Year Total
Benefits				
Revenue uplift	\$2.1M	\$3.8M	\$5.2M	\$11.1M
Cost reduction	\$385K	\$425K	\$450K	\$1.26M
Total Benefits	\$2.485M	\$4.225M	\$5.65M	\$12.36M
Costs				
Development (one-time)	\$140K	—	—	\$140K
Annual operations	\$185K	\$212K	\$239K	\$636K
Total Costs	\$325K	\$212K	\$239K	\$776K
Net Benefit	\$2.16M	\$4.013M	\$5.411M	\$11.584M
ROI	78%	195%	352%	1,494%
Payback Period	10.2 months	—	—	—

5.5 Financial Summary: Conservative Case

Assumptions:

- Revenue uplift: 50% of base case (conservative on personalization impact)
- Cost reduction: 80% of base case (conservative on maintenance reduction)
- Development cost: 120% of base case (schedule risk)
- Operational cost: 110% of base case (higher LLM usage)

Metric	Year 1	Year 2	Year 3	3-Year Total
Benefits				
Revenue uplift	\$1.05M	\$1.9M	\$2.6M	\$5.55M
Cost reduction	\$308K	\$340K	\$360K	\$1.008M
Total Benefits	\$1.358M	\$2.24M	\$2.96M	\$6.558M
Costs				

Development	\$168K	—	—	\$168K
Annual operations	\$204K	\$233K	\$263K	\$700K
Total Costs	\$372K	\$233K	\$263K	\$868K
Net Benefit	\$986K	\$2.007M	\$2.697M	\$5.69M
ROI	36%	97%	204%	656%
Payback Period	16.4 months	—	—	—

5.6 Financial Summary: Optimistic Case

Assumptions:

- Revenue uplift: 150% of base case (strong personalization impact, faster time-to-market)
- Cost reduction: 120% of base case (higher maintenance reduction)
- Development cost: 90% of base case (efficient execution)
- Operational cost: 90% of base case (lower LLM usage)

Metric	Year 1	Year 2	Year 3	3-Year Total
Benefits				
Revenue uplift	\$3.15M	\$5.7M	\$7.8M	\$16.65M
Cost reduction	\$462K	\$510K	\$540K	\$1.512M
Total Benefits	\$3.612M	\$6.21M	\$8.34M	\$18.162M
Costs				
Development	\$126K	—	—	\$126K
Annual operations	\$167K	\$191K	\$215K	\$573K
Total Costs	\$293K	\$191K	\$215K	\$699K
Net Benefit	\$3.319M	\$6.019M	\$8.125M	\$17.463M
ROI	213%	315%	578%	2,497%
Payback Period	5.8 months	—	—	—

5.7 Sensitivity Analysis

Key Assumptions & Sensitivity:

Assumption	Base Case	Range	Impact on Year 1 ROI
Pricing optimization uplift	1-2%	0.5-3%	36% to 120% ROI
Personalization AOV uplift	2-4%	1-6%	24% to 132% ROI
Maintenance reduction	60-70% → 20-30%	50-80% → 30-50%	18% to 138% ROI
Development cost	\$140K	\$100-200K	94% to 62% ROI
Annual operational cost	\$185K	\$150-250K	102% to 54% ROI

Most Sensitive To: Personalization AOV uplift ($\pm 50\%$ assumption = $\pm 54\%$ ROI impact)

Break-Even Analysis:

- If revenue uplift is zero and cost reduction is 50% of estimate: Year 1 ROI = -8% (negative)
- If revenue uplift is 50% of estimate and cost reduction is 50% of estimate: Year 1 ROI = 36% (positive)
- Payback period extends to 16.4 months in conservative case but remains positive

6. Alternatives Considered

6.1 Alternative 1: Do Nothing (Status Quo)

Description: Continue operating with the current PostgreSQL-based analytics infrastructure.

Pros:

- No development cost
- No operational risk from new systems
- Team familiar with current tools

Cons:

- Continued slow insights (5-30 minute query latency)
- Continued peak-load fragility
- Continued high maintenance burden (60-70% of data team capacity)
- Continued opportunity cost (\$650K-1.15M annually in foregone capabilities)
- Competitive disadvantage vs. real-time analytics competitors
- **3-year cost of inaction: \$7.6M** (as calculated in Section 3.6)

Recommendation: Not viable. The cost of inaction exceeds the cost of investment by 3:1 over three years.

6.2 Alternative 2: Buy a Managed Warehouse (Snowflake/BigQuery)

Description: Migrate to a managed cloud data warehouse (Snowflake, BigQuery, or similar).

Pros:

- Proven technology, industry standard
- Vendor support and SLAs
- Scales to petabyte scale
- Supports advanced analytics (ML, time-series, etc.)
- Reduces operational burden on data team

Cons:

- **Higher operational cost:** \$300-500K annually (vs. \$185K for proposed solution)
- **Longer implementation:** 4-6 months (vs. 4 months for proposed solution)
- **Vendor lock-in:** Significant switching costs
- **Overkill for current workload:** Designed for petabyte-scale; we have 2TB
- **Data egress costs:** Transferring data out of warehouse is expensive
- **LLM integration complexity:** Requires custom integration for natural language queries
- **Learning curve:** Team must learn new SQL dialect, performance tuning, cost optimization

Financial Comparison (Year 1):

Metric	Proposed Solution	Snowflake/BigQuery
Development cost	\$140K	\$150K
Annual operational cost	\$185K	\$400K

Revenue uplift	\$2.1M	\$2.1M (same)
Cost reduction	\$385K	\$300K (less maintenance reduction)
Total Year 1 Cost	\$325K	\$550K
Year 1 ROI	78%	61%
Payback Period	10.2 months	13.1 months

Recommendation: Not recommended. While viable, the proposed solution delivers better ROI, lower operational cost, and faster payback. Snowflake/BigQuery is a reasonable fallback if the proposed solution encounters technical blockers.

6.3 Alternative 3: Incrementally Patch PostgreSQL

Description: Continue using PostgreSQL but implement targeted optimizations (better indexing, query tuning, read replicas for analytics).

Pros:

- Lower development cost (\$50-80K)
- Minimal operational change
- Team familiar with PostgreSQL

Cons:

- **Addresses symptoms, not root cause:** Doesn't separate transactional and analytical workloads
- **Limited scalability:** Still constrained by PostgreSQL's transactional design
- **Ongoing maintenance burden:** Doesn't reduce 60-70% maintenance load
- **Peak-load fragility remains:** Analytics queries still lock transactional resources
- **Limited revenue uplift:** Can't support real-time personalization or sub-minute query latency
- **Estimated revenue uplift: \$200-300K** (vs. \$2.1M for proposed solution)
- **Estimated cost reduction: \$50-100K** (vs. \$385K for proposed solution)

Financial Comparison (Year 1):

Metric	Proposed Solution	Patch PostgreSQL
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Development cost	\$140K	\$70K
Annual operational cost	\$185K	\$750K (unchanged)
Revenue uplift	\$2.1M	\$250K
Cost reduction	\$385K	\$75K
Total Year 1 Benefit	\$2.485M	\$325K
Total Year 1 Cost	\$325K	\$70K
Year 1 ROI	78%	364%

Note: The patch approach appears to have higher ROI because it has lower cost, but it delivers only 13% of the benefit. This is a false economy.

Recommendation: Not recommended. While lower cost, the patch approach delivers only 13% of the benefit and doesn't address the root cause. It's a short-term fix that delays the inevitable need for a real solution.

6.4 Alternative 4: Proposed Solution (Recommended)

Description: Implement a purpose-built analytical platform that separates transactional and analytical workloads.

Pros:

- **Addresses root cause:** Separates transactional and analytical concerns
- **Best ROI:** 78% Year 1 ROI, 10.2-month payback period
- **Scalable:** Can grow with the business
- **Enables future capabilities:** Real-time personalization, predictive analytics, etc.
- **Lower operational cost:** \$185K annually vs. \$400K+ for managed warehouses
- **Faster time-to-market:** 4 months vs. 4-6 months for managed warehouses
- **Flexibility:** Can evolve to managed warehouse later if needed

Cons:

- **Technical execution risk:** Requires skilled agentic engineers
- **Operational risk:** New platform requires monitoring and optimization
- **Vendor risk:** Depends on open-source tools (Dagster, dbt, DuckDB)

Mitigation:

- Use proven, battle-tested open-source tools
- Implement comprehensive monitoring and alerting
- Hire experienced engineers; invest in training
- Establish clear SLAs and escalation procedures

Recommendation: Proceed with implementation. This alternative delivers the best balance of cost, benefit, and risk.

6.5 Summary: Alternatives Comparison

Criterion	Do Nothing	Snowflake/BigQuery	Patch PostgreSQL	Proposed Solution
Financial				
Year 1 ROI	Negative	61%	364%	78%
3-year ROI	Negative	142%	156%	352%
Payback period	Never	13.1 mo	3.2 mo	10.2 mo
Operational				
Implementation time	—	4-6 mo	1-2 mo	4 mo
Operational cost (annual)	\$2.1M	\$400K	\$750K	\$185K
Maintenance burden	60-70%	30-40%	55-65%	20-30%
Strategic				
Addresses root cause	No	Yes	No	Yes
Enables future capabilities	No	Yes	No	Yes
Vendor lock-in	No	High	No	Low
Scalability	Limited	Unlimited	Limited	Good
Recommendation	✗ Not viable	⚠ Fallback	✗ False economy	✓ Proceed

7. Risk Assessment & Mitigation

7.1 Technical Risks

Risk	Probability	Impact	Mitigation
Delayed implementation	Medium	High	Allocate experienced engineers; establish clear milestones; use agile methodology with 2-week sprints
Data quality issues during migration	Medium	Medium	Build comprehensive data validation tests; use synthetic data generator to stress-test before production; implement gradual rollout
Performance issues at scale	Low	High	Conduct thorough load testing; use proven, battle-tested tools; implement monitoring and alerting
Integration issues with transactional DB	Low	Medium	Prototype incremental extraction early; establish clear data contracts; test with production-like data volumes
LLM API reliability	Low	Medium	Implement fallback mechanisms; use multiple LLM providers; establish SLAs with vendors

7.2 Operational Risks

Risk	Probability	Impact	Mitigation
Adoption by business teams	Low	Medium	Provide training and documentation; start with high-impact use cases; build intuitive interfaces
Resistance from data team	Medium	Medium	Involve team in design; emphasize that this frees them from maintenance; provide training and support
Knowledge gaps in new tools	Medium	Medium	Hire experienced engineers; invest in team training; use proven open-source tools with strong communities
Operational incidents in new platform	Medium	Medium	Implement comprehensive monitoring; establish clear escalation procedures; maintain runbooks

7.3 Business Risks

Risk	Probability	Impact	Mitigation
Revenue uplift doesn't materialize	Medium	High	Start with pricing optimization (highest confidence); validate assumptions with A/B testing; phase implementation
Competitive response	High	Low	Execute quickly; focus on operational excellence; build sustainable competitive

			advantage through data culture
Market conditions change	Low	Medium	Build flexibility into architecture; avoid long-term vendor contracts; maintain option to scale back if needed

7.4 Downside Scenario

Pessimistic Case Assumptions:

- Revenue uplift: 25% of base case (only pricing optimization realized)
- Cost reduction: 50% of base case (maintenance reduction slower than expected)
- Development cost: 150% of base case (schedule slippage)
- Operational cost: 120% of base case (higher than expected LLM usage)

Downside Case Financial Impact (Year 1):

Metric	Base Case	Downside Case
Revenue uplift	\$2.1M	\$525K
Cost reduction	\$385K	\$193K
Total benefit	\$2.485M	\$718K
Development cost	\$140K	\$210K
Annual operational cost	\$185K	\$222K
Total cost	\$325K	\$432K
Net benefit	\$2.16M	\$286K
ROI	78%	10%
Payback period	10.2 months	27.4 months

Even in downside case, the initiative remains ROI-positive with 10% Year 1 ROI and payback within 27 months. However, this assumes the initiative is not abandoned midway. If abandoned after 6 months, the sunk cost would be \$70K (half of development cost), and the initiative would be negative.

8. Success Metrics & Governance

8.1 Success Metrics (KPIs)

KPI	Current	Target (Month 6)	Target (Month 12)	Owner	Measurement
Performance					
Query latency (p95)	12 min	2 min	<5 sec	Data Eng Lead	Query monitoring logs
Query latency (p99)	28 min	5 min	<15 sec	Data Eng Lead	Query monitoring logs
Data freshness	24 hr	1 hr	<5 min	Data Eng Lead	Ingestion logs
System availability	97.8%	98.5%	99.5%	Data Eng Lead	Uptime monitoring
Operational					
Incidents per month	2.8	1.5	<0.5	Data Eng Lead	Incident tracking
MTTR	5.2 hr	2 hr	30 min	Data Eng Lead	Incident logs
Maintenance burden	60-70%	40-50%	20-30%	VP Data	Time tracking
Business					
Pricing optimization frequency	1x/day	4x/day	15-min intervals	VP Product	Pricing system logs
AOV (with personalization)	\$85	\$85	\$87-89	VP Product	Transaction data
New capabilities launched	0-1/quarter	1-2/quarter	2-3/quarter	VP Data	Project tracking

User Adoption					
Business users querying data	5-10	15-20	30-40	VP Analytics	Access logs
Queries per week	50-100	200-300	500-800	VP Analytics	Query logs

8.2 Governance & Reporting

Steering Committee: CTO, CFO, VP Data, VP Product, VP Analytics

- **Frequency:** Monthly
- **Agenda:** KPI review, risk assessment, budget tracking, escalation of blockers
- **Accountability:** CTO owns delivery; CFO owns budget; VP Data owns operational readiness

Project Management:

- **Methodology:** Agile (2-week sprints)
- **Tracking:** Jira or equivalent
- **Cadence:** Daily standup (15 min), weekly planning (1 hr), bi-weekly demo (1 hr)
- **Escalation:** Weekly escalation to CTO if blockers not resolved within 48 hours

Financial Tracking:

- **Budget:** Tracked weekly against \$140K development budget
- **Actuals:** Compared to forecast monthly
- **Variance:** Any variance >10% escalated to CFO

9. Implementation Roadmap

9.1 Phase 1: Validation (Weeks 1-2)

Objective: Validate technical approach and business assumptions

Activities:

- Build synthetic data generator with realistic failure modes
- Validate platform architecture with production-like data volumes

- Stress test the entire system
- Conduct proof-of-concept for natural language query interface

Deliverables:

- Synthetic data generator (code + documentation)
- Architecture validation report
- Proof-of-concept demo
- Go/No-Go decision

Owner: Lead Data Engineer + Agentic Engineer

Success Criteria:

- Platform handles 50,000 orders/day without performance degradation
- Synthetic data generator successfully injects 5+ failure modes
- Natural language queries return accurate results 95% of the time

9.2 Phase 2: Development (Weeks 3-8)

Objective: Build the analytical platform

Activities:

- Implement incremental data extraction from transactional database
- Build data transformation layer (cleansing, deduplication, enrichment)
- Implement high-performance analytical engine
- Build natural language query interface
- Implement autonomous monitoring and remediation

Deliverables:

- Fully functional analytical platform
- Comprehensive test suite
- Operational runbooks
- Team training materials

Owner: Lead Data Engineer + Agentic Engineer + Backend Engineer

Success Criteria:

- All functional requirements met (BR-001 through BR-009)
- 95%+ test coverage
- Zero critical bugs in production-like testing

9.3 Phase 3: Pilot (Weeks 9-12)

Objective: Validate business benefits in production-like environment

Activities:

- Deploy to pilot environment
- Run parallel with existing system for 2 weeks
- Train pilot users (5-10 business analysts)
- Collect feedback and iterate
- Validate KPIs

Deliverables:

- Pilot deployment (production-like environment)
- User feedback report
- KPI validation report
- Rollout plan

Owner: VP Data + VP Analytics

Success Criteria:

- Query latency meets targets (p95 <5 sec)
- System availability >99%
- Business users report improved decision velocity
- No critical incidents

9.4 Phase 4: Production Rollout (Weeks 13-16)

Objective: Migrate analytics workloads to new platform

Activities:

- Gradual migration of analytics workloads (week by week)
- Monitor performance and reliability
- Optimize based on real-world usage
- Decommission legacy system

Deliverables:

- Production deployment
- Performance optimization report

- Legacy system decommissioning plan

Owner: Lead Data Engineer

Success Criteria:

- All analytics workloads migrated
- System availability >99.5%
- Cost reduction targets met
- Zero unplanned downtime

9.5 Phase 5: Optimization & Scale (Weeks 17+)

Objective: Optimize for cost and performance; enable advanced use cases

Activities:

- Performance optimization and tuning
- Cost optimization (query patterns, caching, etc.)
- Enable real-time personalization (Phase 2)
- Enable predictive analytics (Phase 2)
- Build organizational analytics capability

Owner: VP Data (ongoing)

10. Stakeholder Alignment

10.1 Executive Leadership

Stakeholder: Chief Executive Officer

Benefit:

- Improved profitability: \$2.1M revenue uplift + \$385K cost reduction = \$2.485M Year 1 benefit
- Reduced operational risk: Autonomous monitoring reduces incidents and MTTR
- Strategic enablement: Real-time personalization and predictive analytics become possible

Commitment Required:

- Approve \$140K development budget
- Approve \$185K annual operational budget

- Support 4-month implementation timeline
- Allocate 3 FTE for 4 months

Sign-Off: _____ **Date:** _____

10.2 Chief Financial Officer

Stakeholder: Chief Financial Officer

Benefit:

- Positive ROI: 78% Year 1 ROI, 10.2-month payback period
- Cost reduction: \$385K annually in operational savings
- Improved cash flow: Shift from CapEx to OpEx (operational expense)

Commitment Required:

- Approve \$140K development budget (one-time)
- Approve \$185K annual operational budget
- Allocate budget for LLM API costs (\$34K Year 1)
- Track actual costs vs. forecast monthly

Sign-Off: _____ **Date:** _____

10.3 Chief Technology Officer

Stakeholder: Chief Technology Officer

Benefit:

- Modernized technology stack
- Reduced technical debt
- Improved team morale (less maintenance, more innovation)
- Competitive advantage through data capabilities

Commitment Required:

- Own delivery and accountability
- Allocate 3 FTE for 4 months
- Hire or contract experienced agentic engineers
- Establish clear SLAs and escalation procedures

Sign-Off: _____ **Date:** _____

10.4 VP of Data & Analytics

Stakeholder: VP of Data & Analytics

Benefit:

- Reduced maintenance burden: 60-70% → 20-30% of team capacity
- Faster time-to-market for new capabilities
- Improved system reliability (fewer incidents)
- Opportunity to build advanced analytics capabilities

Commitment Required:

- Participate in design and requirements gathering
- Provide 1 FTE for 4 months (Lead Data Engineer)
- Invest in team training
- Establish KPIs and success metrics

Sign-Off: _____ **Date:** _____

10.5 VP of Product

Stakeholder: VP of Product

Benefit:

- Real-time insights for product decisions
- Faster time-to-market for new features
- Ability to A/B test and optimize in real-time
- Foundation for real-time personalization

Commitment Required:

- Participate in design (natural language query interface)
- Identify high-impact use cases for pilot
- Allocate 2-3 business analysts for pilot testing
- Drive adoption of new tools by product teams

Sign-Off: _____ **Date:** _____

10.6 VP of Analytics

Stakeholder: VP of Analytics

Benefit:

- Faster query latency (5-30 min → <5 sec)
- Self-service analytics (reduce dependency on data engineers)
- More reliable system (fewer incidents)
- Foundation for advanced analytics

Commitment Required:

- Participate in design (natural language query interface)
- Lead pilot testing with business analysts
- Provide feedback and iterate
- Train analytics team on new tools

Sign-Off: _____ **Date:** _____

11. Assumptions & Constraints Register

11.1 Key Assumptions

ID	Assumption	Confidence	Impact if Wrong	Mitigation
A-001	Incremental data extraction can reduce data movement by 80-90%	High	Revenue uplift reduced by 20-30%	Prototype early; validate with production data
A-002	Real-time personalization can increase AOV by 2-4%	Medium	Revenue uplift reduced by 50%	A/B test with pilot users; start conservative
A-003	Maintenance burden can be reduced from 60-70% to 20-30%	Medium	Cost reduction reduced by 50%	Implement comprehensive monitoring; establish clear runbooks
A-004	LLM API costs will remain <\$50K annually	Medium	Operational costs increase by 50%+	Monitor usage patterns; implement rate limiting

A-005	Development can be completed in 4 months with 3 FTE	Medium	Implementation delays by 2-3 months	Allocate experienced engineers; use proven tools
A-006	Open-source tools (Dagster, dbt, DuckDB) will remain actively maintained	High	Vendor risk; potential migration cost	Use battle-tested tools; maintain vendor relationships
A-007	Business teams will adopt natural language query interface	Medium	User adoption slower than expected	Invest in training; start with high-impact use cases
A-008	Pricing optimization algorithm will be effective	Medium	Revenue uplift from pricing reduced by 50%	Validate algorithm with historical data; A/B test

11.2 Constraints

ID	Constraint	Impact	Mitigation
C-001	Data team capacity: 8 FTE	Limits parallel work; requires prioritization	Hire contractors for specific tasks; use agile methodology
C-002	Transactional database cannot be modified (production risk)	Limits optimization opportunities; requires read replicas	Use CDC (Change Data Capture) for incremental extraction
C-003	Business continuity: Analytics cannot go down during peak traffic	Requires high availability; limits testing windows	Implement comprehensive testing; use canary deployments
C-004	Regulatory compliance: Data must remain within geographic boundaries	Limits cloud provider options	Evaluate on-premises and private cloud options
C-005	Budget constraints: \$140K development,	Limits scope; requires prioritization	Phase implementation; start

	\$185K annual operations		with MVP
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12. Appendix: Target Architecture (High-Level)

12.1 Architecture Overview

The analytical platform consists of five layers:

Layer 1: Source

- Transactional database (PostgreSQL)
- Data generator (injects realistic data and failure modes)

Layer 2: Ingestion

- Incremental data extraction using Change Data Capture (CDC)
- Orchestrated by a workflow engine

Layer 3: Transformation

- Data cleansing, deduplication, enrichment
- Implemented using declarative transformation language
- Follows Medallion architecture (bronze → silver → gold)

Layer 4: Warehouse

- High-performance analytical engine
- Optimized for OLAP queries
- Supports complex analytical queries

Layer 5: Consumption

- Natural language query interface
- Exposes analytical data to business users
- Supports both structured queries and exploratory analysis

Layer 6: Monitoring

- Autonomous monitoring and remediation
- Detects anomalies and failures
- Proposes and executes fixes

- Generates post-incident reports

12.2 Data Flow

Plain Text

```
PostgreSQL (Transactional)
↓
CDC / Incremental Extraction
↓
Orchestration Layer
↓
Transformation Layer (Bronze → Silver → Gold)
↓
Analytical Warehouse
↓
Natural Language Interface
↓
Business Users (via Claude Desktop, etc.)
↑
Autonomous Monitoring (detects issues, proposes fixes)
```

12.3 Technology Stack (Reference Only)

Note: Technology selection is a technical decision, not a business requirement. The following is provided for reference only.

- **Orchestration:** Workflow engine for reliable, incremental data extraction
- **Transformation:** Declarative transformation language for data modeling
- **Warehouse:** High-performance analytical engine optimized for OLAP queries
- **Natural Language Interface:** API that translates natural language to analytical queries
- **Monitoring:** Multi-agent system for autonomous monitoring and remediation

13. Conclusion

The Analytical Data Platform Initiative addresses a material operational constraint that is impacting revenue, operational efficiency, and competitive positioning. The proposed solution delivers:

- **Positive ROI:** 78% Year 1 ROI, 10.2-month payback period
- **Revenue uplift:** \$2.1M annually (conservative estimate)
- **Cost reduction:** \$385K annually

- **Strategic enablement:** Foundation for real-time personalization, predictive analytics, and data democratization

The cost of inaction (\$7.6M over 3 years) exceeds the cost of investment (\$776K over 3 years) by a factor of 3:1.

Recommendation: Proceed with implementation immediately.

Approval Sign-Off

Role	Name	Signature	Date
Chief Executive Officer	—	—	—
Chief Financial Officer	—	—	—
Chief Technology Officer	—	—	—
VP of Data & Analytics	—	—	—
VP of Product	—	—	—

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